

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2013-14

Paper No. MTB-606: Number Theory

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Show that for any positive integer m : $(ma, mb) = m(a, b)$. [2]
 - (b) Find integers x, y such that $9 = 657x + 963y$. [2]
 - (c) Prove that if n is composite, then it has a prime divisor p satisfying $p \leq \sqrt{n}$. [2]
 - (d) If $2^n - 1$ is prime, prove that n is prime. [2]
 - (e) Show that $ax \equiv ay \pmod{m}$ if and only if $x \equiv y \pmod{\frac{m}{(a, m)}}$. [2]
 - (f) If n is composite and $n > 4$, prove that $(n - 1)! \equiv 0 \pmod{n}$. [2]
 - (g) Find all Pythagorean triples whose terms form an arithmetic progression. [2]
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Question 2.

- (a) If $g = \gcd(a, b)$, prove that there exist integers x and y such that $g = ax + by$. [7]
 - (b) If $m > 0$, prove that $[ma, mb] = m[a, b]$. Also prove that $a, b = |ab|$. [7]
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Question 3.

- (a) Prove that if $(a, m) = (b, m) = 1$, then $(ab, m) = 1$. [6]
 - (b) State and prove the Fundamental Theorem of Arithmetic. [8]
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Question 4.

- (a) State and prove the Chinese Remainder Theorem. [5]
 - (b) Find all integers that have remainders 1 or 2 when divided by each of 3, 4, and 5. [9]
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Question 5.

- (a) State and prove Wilson's theorem. [7]
- (b) Prove that if $(m, n) = 1$, then $\phi(mn) = \phi(m)\phi(n)$. Show that the condition $(m, n) = 1$ is necessary. [7]
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Question 6.

- (a) If p is prime, prove that $x^2 \equiv -1 \pmod{p}$ has a solution if and only if $p = 2$ or $p \equiv 1 \pmod{4}$. [7]
- (b) Show that for $n \geq 1$: $\sum_{d|n} \phi(d) = n$. [7]
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Question 7.

- (a) Let p be an odd prime and let $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p-1} = \frac{a}{b}$ where $\gcd(a, b) = 1$. Prove that $p \mid a$. [7]
- (b) Find all integral solutions of $32x + 6y + 15z = 1$. [7]
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